benefits of economic activity may be unevenly distributed, incentivizing those who benefit most from it to disregard the harms to others. Second, under intense market competition, businesses tend to focus much more on short-term gains than on long-term outcomes. With this mindset, companies often pursue something that can make a lot of profit in the short term, even if it poses a societal risk in the long term. We will now discuss how corporate competitive pressures could play out with AIs and the potential negative impacts.

3.2.1 Economic Competition Undercuts Safety

Competitive pressure is fueling a corporate AI race. To obtain a competitive advantage, companies often race to offer the first products to a market rather than the safest. These dynamics are already playing a role in the rapid development of AI technology. At the launch of Microsoft's AI-powered search engine in February 2023, the company's CEO Satya Nadella said, "A race starts today... we're going to move fast." Only weeks later, the company's chatbot was shown to have threatened to harm users [47]. In an internal email, Sam Schillace, a technology executive at Microsoft, highlighted the urgency in which companies view AI development. He wrote that it would be an "absolutely fatal error in this moment to worry about things that can be fixed later" [48].

Competitive pressures have contributed to major commercial and industrial disasters. Throughout the 1960s, Ford Motor Company faced competition from international car manufacturers as the share of imports in American car purchases steadily rose [49]. Ford developed an ambitious plan to design and manufacture a new car model in only 25 months [50]. The Ford Pinto was delivered to customers ahead of schedule, but with a serious safety problem: the gas tank was located near the rear bumper, and could explode during rear collisions. Numerous fatalities and injuries were caused by the resulting fires when crashes inevitably happened [51]. Ford was sued and a jury found them liable for these deaths and injuries [52]. The verdict, of course, came too late for those who had already lost their lives. As Ford's president at the time was fond of saying, "Safety doesn't sell" [53].

Boeing, aiming to compete with its rival Airbus, sought to deliver an updated, more fuel-efficient model to the market as quickly as possible. The head-to-head rivalry and time pressure led to the introduction of the Maneuvering Characteristics Augmentation System, which was designed to enhance the aircraft's stability. However, inadequate testing and pilot training ultimately resulted in the two fatal crashes only months apart, with 346 people killed [54]. We can imagine a future in which similar pressures lead companies to cut corners and release unsafe AI systems.

A third example is the Bhopal gas tragedy, which is widely considered to be the worst industrial disaster ever to have happened. In December 1984, a vast quantity of toxic gas leaked from a Union Carbide Corporation subsidiary plant manufacturing pesticides in Bhopal, India. Exposure to the gas killed thousands of people and injured up to half a million more. Investigations found that, in the run-up to the disaster, safety standards had fallen significantly, with the company cutting costs by neglecting equipment maintenance and staff training as profitability fell. This is often considered a consequence of competitive pressures [55].

"Nothing can be done at once hastily and prudently." *Publilius Syrus*

Competition incentivizes businesses to deploy potentially unsafe AI systems. In an environment where businesses are rushing to develop and release products, those that follow rigorous safety procedures will be slower and risk being out-competed. Ethically-minded AI developers, who want to proceed more cautiously and slow down, would give more unscrupulous developers an advantage. In trying to survive commercially, even the companies that want to take more care are likely to be swept along by competitive pressures. There may be attempts to implement safety measures, but with more of an emphasis on capabilities than on safety, these may be insufficient. This could lead us to develop highly powerful AIs before we properly understand how to ensure they are safe.

3.2.2 Automated Economy

Corporations will face pressure to replace humans with AIs. As AIs become more capable, they will be able to perform an increasing variety of tasks more quickly, cheaply, and effectively than human workers. Companies will therefore stand to gain a competitive advantage from replacing their employees with AIs. Companies that choose not to adopt AIs would likely be out-competed, just as a clothing company using manual looms would be unable to keep up with those using industrial ones.

AIs could lead to mass unemployment. Economists have long considered the possibility that machines will replace human labor. Nobel Prize winner Wassily Leontief said in 1952 that, as technology advances, "Labor will become less and less important... more and more workers will be replaced by machines" [56]. Previous technologies have augmented the productivity of human labor. AIs, however, could differ profoundly from previous innovations. Advanced AIs capable of automating human labor should be regarded not merely as tools, but as agents. Human-level AI agents would, by definition, be able to do everything a human could do. These AI agents would also have important advantages over human labor. They could work 24 hours a day, be copied many times and run in parallel, and process information much more quickly than a human would. While we do not know when this will occur, it is unwise to discount the possibility that it could be soon. If human labor is replaced by AIs, mass unemployment could dramatically increase inequality, making individuals dependent on the owners of AI systems.

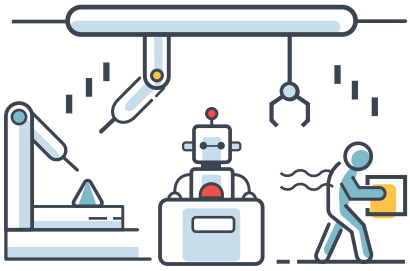

Figure 9: As AIs automate increasingly many tasks, the economy may become largely run by AIs. Eventually, this could lead to human enfeeblement and dependence on AIs for basic needs.

Automated AI R&D. AI agents would have the potential to automate the research and development (R&D) of AI itself. AI is increasingly automating parts of the research process [57], and this could lead to AI capabilities growing at increasing rates, to the point where humans are no longer the driving force behind AI development. If this trend continues unchecked, it could escalate risks associated with AIs progressing faster than our capacity to manage and regulate them. Imagine that we created an AI that writes and thinks at the speed of today's AIs, but that it could also perform world-class AI research. We could then copy that AI and create $10{,}000$ world-class AI researchers that operate at a pace $100\times$ times faster than humans. By automating AI research and development, we might achieve progress equivalent to many decades in just a few months.

Conceding power to AIs could lead to human enfeeblement. Even if we ensure that the many unemployed humans are provided for, we may find ourselves completely reliant on AIs. This would likely emerge not from a violent coup by AIs, but from a gradual slide into dependence. As society's challenges become ever more complex and fast-paced, and as AIs become ever more intelligent and quick-thinking, we may forfeit more and more functions to them out of convenience. In such a state, the only feasible solution to the complexities and challenges compounded by AIs may be to rely even more heavily on AIs. This gradual process could eventually lead to the delegation of nearly all intellectual, and eventually physical, labor to AIs. In such a world, people might have few incentives to gain knowledge and cultivate skills, potentially leading to a state of enfeeblement [58]. Having lost our know-how and our understanding of how civilization works, we would become completely dependent on AIs, a scenario not unlike the one depicted in the film WALL-E. In such a state, humanity is not flourishing and is no longer in effective control.

As we have seen, there are classic game-theoretic dilemmas where individuals and groups face incentives that are incompatible with what would make everyone better off. We see this with a military AI arms race, where the world is made less safe by creating extremely powerful AI weapons, and we see this in a corporate AI race, where an AI's power and development is prioritized over its safety. To address these dilemmas that

give rise to global risks, we will need new coordination mechanisms and institutions. It is our view that failing to coordinate and stop AI races would be the most likely cause of an existential catastrophe.

3.3 Evolutionary Pressures

As discussed above, there are strong pressures to replace humans with AIs, cede more control to them, and reduce human oversight in various settings, despite the potential harms. We can re-frame this as a general trend resulting from evolutionary dynamics, where an unfortunate truth is that AIs will simply be more fit than humans. Extrapolating this pattern of automation, it is likely that we will build an ecosystem of competing AIs over which it may be difficult to maintain control in the long run. We will now discuss how natural selection influences the development of AI systems and why evolution favors selfish behaviors. We will also look at how competition might arise and play out between AIs and humans, and how this could create catastrophic risks. This section draws heavily from "*Natural Selection Favors AIs over Humans*" [59, 60].

Fitter technologies are selected, for good and bad. While most people think of evolution by natural selection as a biological process, its principles shape much more. According to the evolutionary biologist Richard Lewontin [61], evolution by natural selection will take hold in any environment where three conditions are present: 1) there are differences between individuals; 2) characteristics are passed onto future generations and; 3) the different variants propagate at different rates. These conditions apply to various technologies.

Consider the content-recommendation algorithms used by streaming services and social media platforms. When a particularly addictive content format or algorithm hooks users, it results in higher screen time and engagement. This more effective content format or algorithm is consequently "selected" and further fine-tuned, while formats and algorithms that fail to capture attention are discontinued. These competitive pressures foster a "survival of the most addictive" dynamic. Platforms that refuse to use addictive formats and algorithms become less influential or are simply outcompeted by platforms that do, leading competitors to undermine wellbeing and cause massive harm to society [62].

The conditions for natural selection apply to AIs. There will be many different AI developers who make many different AI systems with varying features and capabilities, and competition between them will determine which characteristics become more common. Second, the most successful AIs today are already being used as a basis for their developers' next generation of models, as well as being imitated by rival companies. Third, factors determining which AIs propagate the most may include their ability to act autonomously, automate labor, or reduce the chance of their own deactivation.

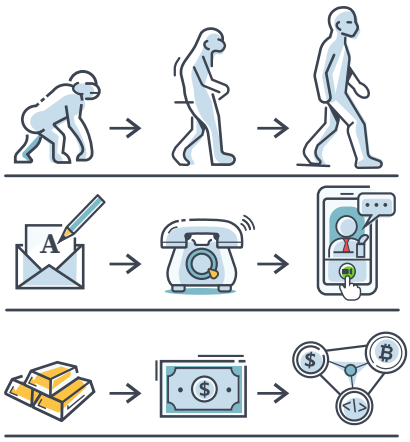

Figure 10: Evolutionary processes are not limited to the realm of biology.

Natural selection often favors selfish characteristics. Natural selection influences which AIs propagate most widely. From biological systems, we see that natural selection often gives rise to selfish behaviors that promote one's own genetic information: chimps attack other communities [63], lions engage in infanticide [64], viruses evolve new surface proteins to deceive and bypass defense barriers [65], humans engage in nepotism, some ants enslave others [66], and so on. In the natural world, selfishness often emerges as a dominant strategy; those that prioritize themselves and those similar to them are usually more likely to survive, so these traits become more prevalent. Amoral competition can select for traits that we think are immoral.

Examples of selfish behaviors. For concreteness, we now describe many selfish traits—traits that expand AIs' influence at the expense of humans. AIs that automate a task and leave many humans jobless have engaged in selfish behavior; these AIs may not even be aware of what a human is but still behave selfishly towards them—selfish behaviors do not require malicious intent. Likewise, AI managers may engage in selfish and "ruthless" behavior by laying off thousands of workers; such AIs may not even believe they did anything